

Project Design Phase

Problem – Solution Fit Template

Date	2 July 2026
Team ID	24A55A0518
Project Name	Metro Ticket Generating System
Maximum Marks	2 Marks

Solution Fit:

The Metro Ticket Generating System, developed on the ServiceNow platform, provides a digital and automated solution to overcome the limitations of traditional metro ticketing systems. The application allows passengers to book metro tickets online by selecting the source station, destination station, number of passengers, journey type, and payment mode through a user-friendly Service Portal interface.

The system automatically calculates the fare based on predefined metro routes, reducing manual effort and errors. After booking, it generates a unique ticket number, creates a digital ticket record, and produces a QR code for secure and contactless ticket verification. The booking details are stored in a centralized Metro Database for efficient management and retrieval.

Additionally, administrators can manage metro stations, fare information, and ticket records through the ServiceNow platform. By digitizing the complete ticketing workflow, the solution reduces waiting time, improves operational efficiency, enhances passenger experience, and supports secure and scalable metro transportation services.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? - Daily metro commuters - Occasional metro passengers - Metro ticket validators / staff - Metro administration personnel - ServiceNow system administrators	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices? - Long queues and waiting time at counters - Manual ticketing process causes delays - Limited ticket booking channels - Risk of losing physical tickets - Lack of centralized records & transparency - Dependence on manual validation	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? - Physical ticket counters - Smart card based ticketing - Manual ticket validation - Existing mobile ticketing applications - Paper based tickets	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? - Book metro tickets quickly - Avoid long queues at ticket counters - Calculate fares accurately - Generate digital tickets instantly - Validate tickets using QR code - Manage ticket records efficiently - Ensure secure and contactless travel	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? - Dependence on manual and paper based processes - Lack of automation in ticket booking & validation - Absence of centralized digital system - Inefficient data management - Limited use of digital technologies - No integration between booking, payment, and validation processes	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? - Wait in queues and buy physical tickets - Use smart cards - Prefer mobile payment options - Check fares and routes before travel - Show physical ticket for validation - Rely on staff for ticket related queries	
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. - Long waiting times at ticket counters - Increasing demand for contactless travel - Growth in digital payment usage - Need for faster and convenient travel - Government push for digital transformation	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits really. If you are working on a new business proposition, then keep it blank: until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Metro Ticket Generating System - Online metro ticket booking through Service Portal - Automatic fare calculation based on source & destination - Unique ticket ID generation - QR code ticket generation for secure travel - QR code ticket validation at entry/exit - Centralized metro database for tickets & stations - Workflow automation using ServiceNow Flow Designer - Notifications and real-time status updates - Digital, contactless and secure ticketing experience	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Service Portal (Web) - Mobile devices (Responsive) - Email notifications - QR code scanning devices 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Metro counters (for support) - Validator staff for assistance - Helpdesk / Customer support	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? Before: Frustrated, stressed, confused, time consuming, uncertain After: Satisfied, convenient, confident, happy, fast and easy travel			